

CS1005 – Discrete Structures

Quiz 1

Wednesday, 4 February 2025

Time: 10 mins

Marks: 10

Note: Please write in blue or black ink. Use of pencils is not allowed. No articles (calculators etc.) can be exchanged.

1. Which of the following sentences are propositions? Also, explain why.

- Ramzan starts on the sighting of the new moon.
- $x + 1$ is a prime number.
- You may start the quiz.
- Is it raining now?
- Snow falls in summer.

(2.5 Marks,
0.5 each)

0 Marks
w/o explanation

- Sol**
- Yes, since it is either true or false.
 - No, since x is a variable/unknown.
 - No, this is a command.
 - No, this is a question.
 - Yes, it is either true or false.

2. Prove or disprove that $\alpha \leftrightarrow \beta \equiv (\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$. (4.5 Marks)

Sol

α	β	$\alpha \rightarrow \beta$	$\beta \rightarrow \alpha$	$(\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$	$(\alpha \leftrightarrow \beta)$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

Hence, $\alpha \leftrightarrow \beta \equiv (\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$

3. Explain if the following argument is valid or invalid. Use propositional logic. (3 Marks)
- Every dog is a reptile.
 - Every reptile is cold-blooded.
 - Therefore, every dog is cold-blooded.

Sol

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Let P : It is a dog.

Let Q : It is a reptile.

Let R : It is cold blooded.

(a) $P \rightarrow Q$

(b) $Q \rightarrow R$

(c) $\therefore P \rightarrow R$ (Using transitivity rule)

Hence, it is valid